

Project Report

ENG20009

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Abstract

This report is a reflection on the completion of an Arduino-based project with an emphasis on collaborative effort and individual contributions. The report analyses specific parts of the team's pseudocode developed by me. A risk assessment and thorough knowledge of the project and the Arduino platform.

Pseudocode

Address Query

```
Function SDI12_Command(input)
    // Function to check for SDI-12 Commands input to the serial debug monitor

    // Convert device address to string
    address = String(deviceAddress)

    If String(input.charAt(0)) == "?" Then // Checking for ?! - Address Query command
        If input.length() == 1 Then
            SDI12_Output(address) // Return current device address
        Else
            SDI12_Output("ERROR: Invalid Syntax for Address Query command") // Provide
error message if syntax for address query command is wrong
        End If
    End If
End Function
```

Explanation

1. **SDI12_Command(input):**

- Converts the device address to a string.
- Checks if the first character of the input is "?".
- If the input length is 1 (valid address query command), outputs the device address.
- Otherwise, outputs an error message indicating invalid syntax.

Idle Menu

```
Function idle_Menu()  
    // Function to display an Idle Menu when logged out or starting up  
    tft.setTextSize(3)  
    tft.setTextColor(ST77XX_GREEN, ST77XX_BLACK)  
    tft.setCursor(0, 4)  
    tft.println("\n SDI12\n Project")  
    tft.setTextSize(1)  
    tft.print("\n")  
    tft.setTextColor(ST77XX_WHITE, ST77XX_BLACK)  
    tft.println(" In Home Screen \n\n Press (PB2) to Enter")  
End Function
```

Display Menu

```
Function displayMenu(title, options[], numOptions, highlightIndex)  
    tft.setTextSize(2)  
    tft.setTextColor(ST77XX_GREEN, ST77XX_BLACK)  
    tft.setCursor(15, 10)  
    tft.println(title)  
    tft.setTextColor(ST77XX_WHITE, ST77XX_BLACK)  
  
    For i = 0 to numOptions - 1  
        If i == highlightIndex Then  
            tft.setTextColor(ST77XX_BLACK, ST77XX_WHITE) // Highlighted option  
        Else  
            tft.setTextColor(ST77XX_WHITE, ST77XX_BLACK) // Normal option  
        End If  
        tft.println(options[i])  
    End For  
End Function
```

Main Menu

```
Function main_Menu()  
    // Function to display the main menu once logged in and used to select menu options  
    options[] = { " 1. Sensors \n", " 2. SD Card \n", " 3. Return " }  
    displayMenu(" Main Menu \n", options, 3, selection[0] - 1)  
End Function
```

Sensor Menu

```
Function sensor_Menu()  
  // Function to display the sensor menu to select sensor data to be displayed  
  options[] = { " 1. Temperature \n", " 2. Pressure \n", " 3. Humidity \n", " 4. Gas Quality  
\n", " 5. Light Intensity " }  
  displayMenu("Select Sensor", options, 5, selection[1] - 1)  
End Function  
````
```

## Explanation

### 1. **idle\_Menu():**

- Sets text size and color.
- Positions the cursor and prints the idle menu.

### 2. **displayMenu(title, options[], numOptions, highlightIndex):**

- Sets text size and color.
- Prints the title.
- Iterates through the options array.
- Highlights the selected option based on `highlightIndex`.

### 3. **main\_Menu():**

- Calls `displayMenu` with the main menu title and options.

### 4. **sensor\_Menu():**

- Calls `displayMenu` with the sensor menu title and options.

## Reason for this risk assessment

- New task
- New information
- Change to existing work environment
- Report of injury
- Cyclic review
- Other

# Identification of a health and safety hazard

The core focus of this unit is the development of an SDI-12 sensor and data logger using a project-based learning approach. Students are divided into groups of five, each with specific roles. The project involves selecting sensor pairs (temperature, humidity, pressure, gas, light intensity) and integrating them with an Arduino microcontroller using I2C or UART communication protocols. The Arduino is programmed to efficiently acquire and interpret SDI-12 commands. Evaluation includes using UART to SDI-12 and SDI-12 to USB converters. A self-contained device records sensor data and real-time clock information onto an SD card. The system also features a user-friendly graphical interface with an LCD display for sensor selection and data visualization.

## **Hazards to consider:**

### **1. Sensor Integrity:**

The BME680 and BH1750 sensors are delicate; mishandling can damage them.

### **2. Incorrect Wiring:**

Improper wiring can cause short circuits or damage the sensors.

### **3. Software Failure:**

Software malfunction can result in inaccurate measurements, false alarms, or failure to detect critical incidents.

### **4. Personal Safety:**

Adherence to appropriate attire, such as wearing closed-toe shoes, is essential for safety during lab sessions.

### **5. Electrical Hazard:**

Potential electrical hazards include short circuits and electrical shocks when sensors are connected to a power supply.

**6. Technical Complexity:** Calibration, data interpretation, and system integration of the sensors are technically challenging.

### **7. Noise Pollution:**

Excessive buzzer sounds can cause auditory discomfort or harm.

### **8. Project Management:**

Poor project management can lead to delays, budget overruns, and communication breakdowns.

### **9. Data Loss:**

SD card corruption can result in the loss of crucial sensor data.

### **10. Risk of Eye Damage:**

High-intensity light from the BH1750 sensor can potentially cause eye damage.

#### 11. Water Damage:

Sensors are vulnerable to water damage, leading to malfunction and inaccurate readings.

## **Risk rating – risk matrix and definitions**

#### Consequences:

- Minor: Injury requiring minimal medical treatment or first aid.
- Disruptive: Injury requiring medical treatment and sustained lost time.
- Significant: Long-term injuries/disability, short-term hospitalization.
- Critical: Single fatality, severe irreversible disabilities.
- Catastrophic: Multiple fatalities, significant irreversible disabilities.

#### Likelihood:

- Almost Certain: 99% probability, could occur within days to months.
- Likely: >50% probability, could occur within months to years.
- Possible: >10% probability, may occur shortly but might not.
- Unlikely: >1% probability, may occur but not anticipated.
- Rare: <1% probability, requires exceptional circumstances.

#### Risk rating priority:

- High/Very High: Immediate action required.
- Major: Action within 1 to 3 months.
- Moderate: Action within 3 to 6 months.
- Low: Monitor and review ongoing.

# Identify hazards and associated risk scores and controls

## Risk Scoring System:

- **Likelihood:**
  - Almost Certain (5)
  - Likely (4)
  - Possible (3)
  - Unlikely (2)
  - Rare (1)
- **Consequence:**
  - Catastrophic (5)
  - Critical (4)
  - Significant (3)
  - Disruptive (2)
  - Minor (1)
- **Risk Score:** The risk score is calculated by multiplying the likelihood and consequence values. For example, a "Likely" likelihood (4) and a "Significant" consequence (3) would result in a risk score of 12.

The identified hazards in this project include:

### 1. Sensor Integrity

- Who can get harmed and how?: Measurements can be inaccurate due to mishandling.
- Uncontrolled Risk Score: Likely-Significant (4-3) = 12
- Controls required: Protocols for proper handling and storage, training session.
- Residual Risk Score: Possible-Disruptive (3-2) = 6
- Implementation of controls: Training on proper handling.
- Person responsible: All group members

### 2. Personal Safety

- Who can get harmed and how?: Physical injury due to improper attire.
- Uncontrolled Risk Score: Possible-Minor (3-1) = 3
- Controls required: Adherence to attire regulations.
- Residual Risk Score: Unlikely-Minor (2-1) = 2
- Implementation of controls: Guidance on laboratory safety protocols and dress code.
- Person responsible: All group members

### 3.Noise Pollution

- Who can get harmed and how?: Auditory discomfort from buzzer.
- Uncontrolled Risk Score: Likely-Disruptive (3-2) = 8
- Controls required: Instructions on maintaining noise levels, sound attenuation.
- Residual Risk Score: Unlikely-Minor (2-1) = 2
- Implementation of controls: Measures to reduce buzzer sound intensity.
- Person/s responsible: All group members

### 4. Risk of Eye Damage

- Who can get harmed and how?: Eye damage from light sensor.
- Uncontrolled Risk Score: Possible-Significant (3-3) = 9
- Controls required: Use of appropriate eye protection.
- Residual Risk Score: Unlikely-Disruptive (2-1) = 4
- Implementation of controls: Provision of safety eyewear.
- Person responsible: All group members

### 5. Incorrect Wiring

- Who can get harmed and how?: Short circuits from improper installation.
- Uncontrolled Risk Score: Likely-Critical (4-3) = 16
- Controls required: Development of wiring schematic, instructional sessions.
- Residual Risk Score: Possible-Disruptive (3-2) = 6
- Implementation of controls: Comprehensive wiring methodologies.
- Person responsible: All group members

### 6. Project Management Risks

- Who can get harmed and how?: Project delays due to poor management.
- Uncontrolled Risk Score: Possible-Significant (3-3) = 9
- Controls required: Thorough documentation, instructional materials.
- Residual Risk Score: Possible-Minor (3-1) = 3
- Implementation of controls: Technical manual and training sessions.
- Person responsible: All group members

### 7. Data Loss

- Who can get harmed and how?: Challenges in data retrieval.
- Uncontrolled Risk Score: Likely-Significant (4-3) = 12
- Controls required: Comprehensive data backup and recovery strategy.
- Residual Risk Score: Possible-Disruptive (3-2) = 6
- Implementation of controls: Robust data backup scheme.
- Person responsible: All group members

### 8. Electrical Hazard

- Who can get harmed and how?: Electrical hazards from sensor power supply.



- Uncontrolled Risk Score: Likely-Disruptive (4-2) = 8
- Controls required: Adherence to proper connection guidelines, tutor supervision.
- Residual Risk Score: Possible-Disruptive (3-2) = 6
- Implementation of controls: Proper guidance and supervision during initial connection.
- Person responsible: All group members

#### 9. Technical Complexity

- Who can get harmed and how?: Technical challenges in sensor configuration.
- Uncontrolled Risk Score: Likely-Critical (4-4) = 16
- Controls required: Thorough documentation, instructional materials.
- Residual Risk Score: Possible-Significant (3-3) = 9
- Implementation of controls: Technical manual and training sessions.
- Person responsible: All group members

## Reflective Understanding of Engineering Enquiry Project

Throughout the unit, seminars, facilitation sessions, labs, and workshops provided valuable opportunities to acquire and apply essential knowledge for our Arduino-based project. This section reflects on the depth of reflection, evaluation, and interpretation of knowledge received, demonstrating a complete and extensive understanding of the Arduino platform and its applications.

### Seminars:

The seminars provided a foundational understanding of the unit, introducing key concepts related to the Arduino-based project. These sessions offered an overview of the unit objectives, emphasized effective teamwork, and highlighted the role of components such as GPIO, ADC, memory, interrupts, and programming style in embedded system development. Reflecting on these seminars, I gained a deep understanding of the project scope and its technical requirements, enabling effective planning and execution.

### Facilitation Sessions:

These sessions were critical in guiding us through the practical aspects of the Arduino project. They included step-by-step instructions on fundamental programming, GPIO configuration, interfacing with external devices, memory management, and advanced capabilities such as interrupts. By actively engaging in these sessions, I gained hands-on experience with various functionalities on Arduino, solidifying my understanding of the hardware and software involved in designing embedded systems.

**Laboratory Classes:**

The lab classes provided a structured environment to apply theoretical knowledge gained in seminars and facilitation sessions. Through hands-on experimentation, we engaged in activities such as GPIO control, interfacing with ADCs, memory management, and implementing interrupts. These exercises allowed me to solidify my understanding of the Arduino platform and its capabilities. The labs also provided opportunities for teamwork, collaboration, and troubleshooting, enhancing our project skills.

**Workshops:**

The workshops focused on areas of project development such as firmware/software development, prototyping, testing, and project demonstration. These sessions delved into topics like SDI-12 connectivity, advanced interrupts, programming style, and deploying standalone embedded systems. By actively participating in these sessions, I gained extensive knowledge about project design, development, and testing, which helped fine-tune our project strategy and ensure the delivery of a functioning, well-designed product. The use of Wokwi simulation was particularly beneficial, enabling everyone to design and code without any hassle.

## **Communication with Teams and Stakeholders**

Clear and constructive communication was established and maintained with both team members and stakeholders throughout the project. As the team coordinator, I played a pivotal role in facilitating effective communication within the team and between the team and stakeholders. Various verbal, written, and technological approaches were employed to enhance project outcomes. During team meetings, my active engagement can be observed, sharing insightful perspectives and attentively listening to others' ideas. Concise and well-articulated written communication, including emails and documentation, effectively conveyed important project updates and information to keep everyone informed. Furthermore, relevant technological tools were utilized, such as collaborative platforms (Microsoft Teams, WhatsApp, Zoom), to facilitate seamless code and document sharing. By employing these communication strategies, efficient information exchange was ensured, fostering a cohesive and productive work environment. In addition, the team held two physical meetings each week, on Mondays and Tuesdays, to discuss progress and solve potential problems. As a team member and the coordinator, I focused on maintaining communication with team members and stakeholders to ensure project success. Communication is the most important aspect of a team-based project.

# Conclusion

In conclusion, the Arduino-based project completed in the Engineering Enquiry unit was a success. I obtained a thorough understanding of Arduino and successfully applied my knowledge to the project. As team coordinator, I actively contributed to the codebase, troubleshooted sensors, and managed team activities. Despite problems such as a team member's absence, we overcame them through dedication and collaboration. Clear communication tactics allowed seamless teamwork. Overall, this experience improved my skills in embedded system development, teamwork, and communication.